Chapter 2. Computer System Performance

- time
- memory / circuits
- power
- dependability – IEEE Trans on Dependable and Secure Computing 2004/2005 Algirdas Avižienis

2.1 Taxonomy

- 3 classes of applications for computers

① Desktop computers / laptop

Def Designed for single users, normally executes 3rd party SW

≃ 40 years old → driver of progress in computer design

② Servers

– mainframes, minicomputers, supercomputers

Def Designed for multiple users, 3rd party SW, accessed through a network

– scientific applications : simulations
– many simple applications : WEB searches

} large workloads

Supercomputer ex.

– 100,000 processors / cores
– TiBs of memory
– cost 100,000, 1000 000s of $

E.g., server app – protein folding

10^3 ... 2^{10} 1Ki
10^6 ... 2^{20} 1Mi
10^9 ... 2^{30} 1Gi
10^{12} ... 2^{40} 1Ti tera
10^{15} ... 2^{50} 1Pi peta
10^{18} ... 2^{60} 1Ei exa
10^{21} ... 2^{70} 1Zi zetta
10^{23} ... 2^{80} 1Yi yotta

③ <u>Embedded Computers</u> : Computers as Components

- <u>single applications</u>, only 3rd party SW
- cars, phones, TV, game console, avionics, medical apps (e.g., <u>pacemaker</u>).

CPS – cyber physical system

IoT – Internet of Things (IoMT)

Deadlines → performance

Dependability

2.2 Performance metrics

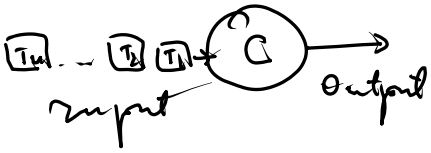
Throughput Vs. Response time

Do the following changes to a computer system <u>increase throughput</u>, <u>decrease response time</u>, or both?

① Replacing the processor with a faster version

⇒ decrease response time

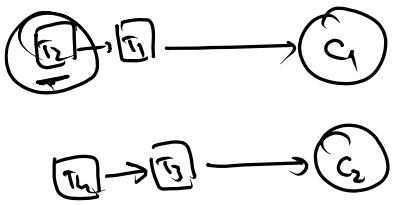
⇒ increase throughput



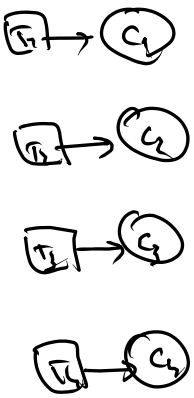
② Adding more processors to a system that uses multiple processors for separate tasks

⇒ increase throughput

⇒ decrease response time



Scenario 1



Scenario 2

$$\text{Performance}_X = \frac{1}{\text{Execution time}_X}$$

$$\text{Performance}_X > \text{Performance}_Y \Rightarrow \boxed{\text{Execution time}_X < \text{Execution time}_Y}$$

Computer X is n times more performant than computer Y

$$\Rightarrow \boxed{\frac{\text{Performance}_X}{\text{Performance}_Y} = n = \frac{\text{Execution time}_Y}{\text{Execution time}_X}}$$

Example

$$\frac{\text{Performance}_A}{\text{Performance}_B} = \frac{\text{Execution time}_B}{\text{Execution time}_A} = \frac{15}{10} = 1.5$$

If computer A runs a program in 10 seconds and computer B runs the same program in 15 seconds

$\Rightarrow$ computer A is 1.5 times better than computer B

Airplane	Passenger capacity	Cruising range (miles)	Cruising speed (m.p.h.)	Passenger throughput (passengers × m.p.h.)
Boeing 777	375	4630	610	228,750
Boeing 747	470	4150	610	286,700
BAC/Sud Concorde	132	4000	1350	178,200
Douglas DC-8-50	146	8720	544	79,424

FIGURE 1.13 The capacity, range, and speed for a number of commercial airplanes. The last column shows the rate at which the airplane transports passengers, which is the capacity times the cruising speed (ignoring range and takeoff and landing times).